```
import os
import openai
import secret
import requests
openai.api_key='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'

# WRITE YOUR CODE HERE
response = openai.Image.create(
  prompt="a dog smoking a cigar in a poker room.",
  n=1,
  size="256x256"
)

print(response)
image_url = response['data'][0]['url']
print(image_url)
```


Text to Image

Dall-E

OpenAI's **DALL·E** text to image is a system that can generate images from textual descriptions, using a deep learning-based approach. The system is trained on a large dataset of text-image pairs, and can generate new images that are similar to the ones in the training data. The system can generate images that are realistic and editable, and can even create images from scratch, without any training data.

DALL·E models can easily generate images from text, making it a great choice for those who want to create quick and easy visual content.

The Images API provides three methods for interacting with images:

1. Creating images from scratch based on a text prompt
2. Creating edits of an existing image based on a new text prompt
3. Creating variations of an existing image

Generating Image

In order to generate an image using the DALL·E we can use the `openai.Image.create` call. Here is an example below:

```
response = openai.Image.create(
  prompt="a dog smoking a cigar in a poker room.",
  n=1,
  size="256x256"
)
```

TRY IT!

You can create an original image based on a text prompt using the image generations endpoint. Images can be sized at 256x256, 512x512, or 1024x1024 pixels, with smaller sizes generate faster. You can request 1-10 images at a time by using the n parameter. A 256x256 would generate faster than 1024x1024. For this course, we will tend to just use the 256 or the 512. These 2 are cheaper and faster to generate. API endpoints are the points where requests are made and responses are sent back. Let's print the content of the response to see what was generated by the API call.

```
print(response)
```

TRY IT!

When we printed the response we see there 3 distinct words `created`, `data` and inside `data` we see a URL. Here we want to focus on `URL`. The URL is the link to the image that the AI generated for us. Data represents all the info generated for the user and creation is pretty much an id. Now that we know what our response looks like we can trim down the response to what we actually want, which is the URL. Remove the `print(response)` line, and add the following.

```
image_url = response['data'][0]['url']
print(image_url)
```

TRY IT!

We can open another tab in our browser, then we can copy and paste the URL to see the image that we have generated.

Try giving it a different size from the standard size mentioned above. For example, 255x255 or 256x1024.

TRY IT!

It should generate an error, please revert back to a 256x256 or 512x512. We want it to be able to generate faster for future outputs. A Try it button is provided below to make sure we are no longer getting errors. It should just generate an URL here.

TRY IT!

Which method would you use to generate an image using DALL-E?

- (●) `openai.Image.create`
- ( ) `openai.Image.generate`
- ( ) `openai.Text.create`
- ( ) `openai.Image.process`

We have to use `openai.Image.create` here is a sample code:

```
response = openai.Image.create(
  prompt="a dog smoking a cigar in a poker room.",
  n=1,
  size="256x256"
)
```

Check It!

Next